

Who wins a hash collision?

A measurement of the collision argument in instant-NGP

measured by `scripts/collision_audit.py` in `ngp-demo`

August 26, 2026

The claim under test

Müller et al. (2022), §3, explain why a hash table with no collision handling still reconstructs a scene faithfully:

“When training samples collide in this way, their gradients average. Consider that the importance to the final reconstruction of such samples is rarely equal. ... As a result, the gradients of the more important samples dominate the collision average and the aliased table entry will naturally be optimized in such a way that it reflects the needs of the higher-weighted point.”

Two distant points that hash to the same row do not overwrite one another. The row is one parameter receiving the *sum* of two pulls, and it settles where the stronger pull wants it. In a radiance field “stronger” means density $\times$ visibility. The question this note answers is whether that story survives being measured, and how much the loser actually pays.

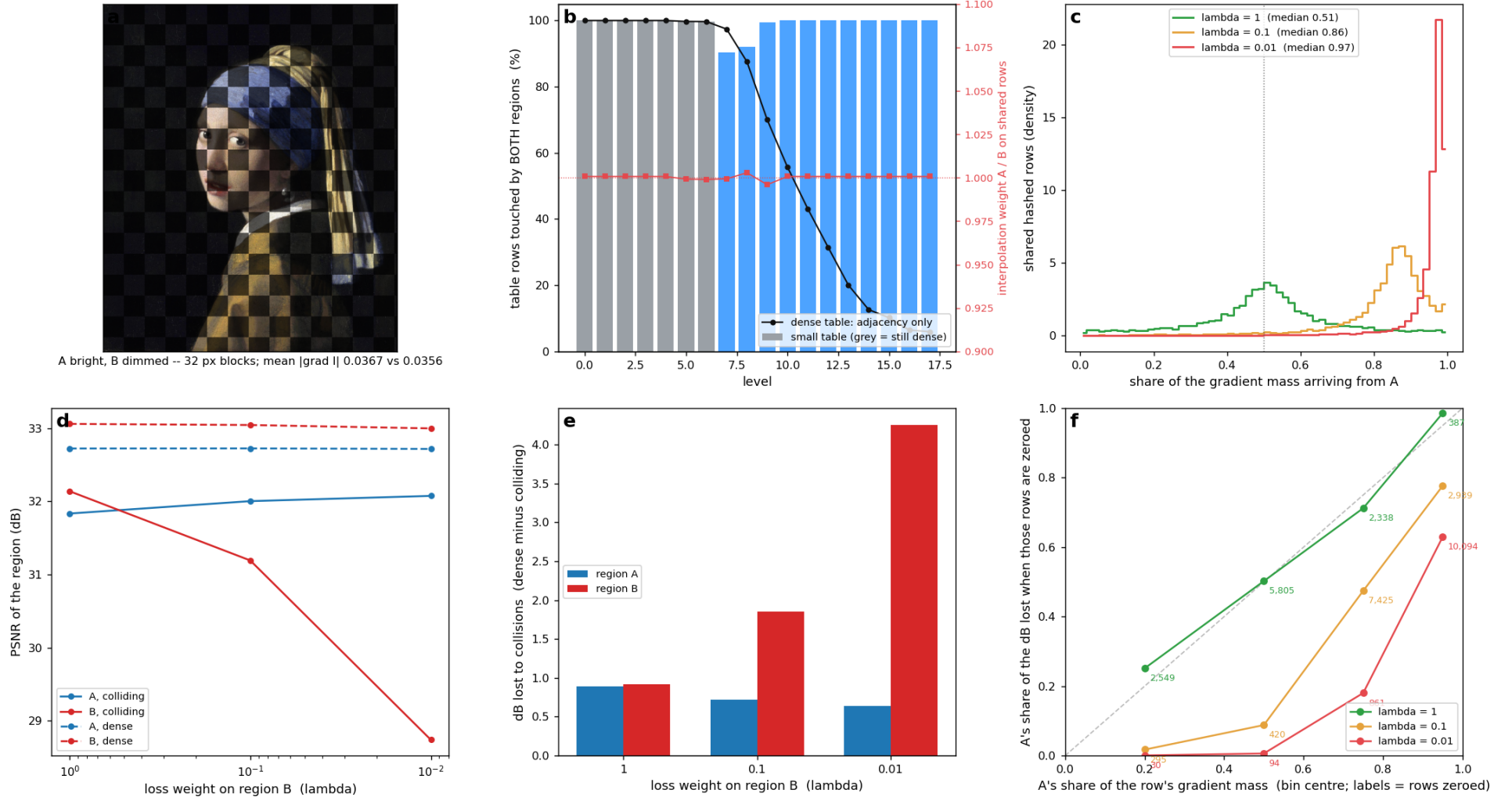
The design, and why it is not circular

The tempting experiment — fit a painting and observe that the detailed parts come out sharper than the flat parts — proves nothing. Detailed regions are sampled against more structure, occupy different levels and hold different amounts of error, so any asymmetry has half a dozen possible causes.

So importance is made an explicit knob and everything else is held fixed:

- The painting is cut into a **checkerboard of 32 px blocks**. Region A is the bright squares of panel (a), region B the dimmed ones. They are the same picture at the same scales, interleaved: 120,728 against 120,640 pixels, mean $|\nabla I|$ 0.0367 against 0.0356. Panel (b) confirms the consequence that matters — on every hashed level the two regions put the *same* interpolation weight on the rows they share, within 0.4%.
- Region B is fitted at **loss weight** λ ; region A always at 1. This is the paper’s “importance”, made a dial: λ multiplies exactly the gradient B delivers, and nothing else.
- Each λ is run **twice**: once with a table small enough that the fine levels collide ($T = 2^{10}$, 11 of 18 levels hashed) and once with a table large enough that **every** level is dense ($T = 2^{19}$, 0 hashed). The dense run is the control that separates *B learns less because its gradients are small* from *B loses its shared rows to A*.

The encoder is the one `scripts/gui_image.py` fits by default (L=18, F=2, coarsest 4 cells, $b=1.35$, finest capped at the pixel count), on the painting at 452×534 , 1,000 steps of Adam at lr 10^{-2} with batch 65,536.



Instant-NGP collision audit: two interleaved regions of one painting, region B fitted at loss weight λ . Every number measured by `scripts/collision_audit.py`.

Figure 1: The audit. **(a)** the two regions. **(b)** how the two regions come to share a table row, level by level: the black line is the dense table, where sharing can only mean adjacency, and it falls to 6.0% at the finest level; the bars are the small table, where sharing is 100.0%. The red squares are the balance check. **(c)** where a shared row's gradient mass comes from, one histogram per λ . **(d)** the outcome: solid = colliding, dashed = dense. **(e)** the same, as what each region lost to collisions. **(f)** zeroing shared rows, grouped by A's share of their gradient mass; the dashed diagonal is "damage in proportion to mass".

What was measured

λ	colliding, $T = 2^{10}$		dense, $T = 2^{19}$		B's loss to collisions
	A (dB)	B (dB)	A (dB)	B (dB)	
1	31.84	32.14	32.72	33.06	0.92 dB
0.1	32.00	31.19	32.73	33.05	1.85 dB
0.01	32.08	28.74	32.72	33.00	4.26 dB

Table 1: Per-region PSNR. 12,692 table rows against 754,783.

- 1. Equal weights, equal outcome.** At $\lambda = 1$ the two regions land -0.30 dB apart with collisions and -0.34 dB apart without, and collisions cost them 0.89 and 0.92 dB respectively. The checkerboard is fair, and a shared table is not by itself unfair.
- 2. Weight alone costs nothing.** Down-weighting B by $100\times$ in the *dense* table moves it by 0.06 dB ($33.06 \rightarrow 33.00$). With its own rows, a region fitted at $\lambda = 0.01$ is fitted just as well; the gradients are smaller but they point the same way, and Adam is scale-free per parameter. This is the control that makes the next line mean something.
- 3. Weight decides who owns a shared row.** The gradient mass arriving at a shared hashed row is split 0.507 in favour of A at $\lambda = 1$ (median over 11,079 rows) — the checkerboard’s balance, showing up in the training signal. At $\lambda = 0.1$ that median is 0.860, at $\lambda = 0.01$ it is 0.970, with 91.1% of shared rows above 0.9. Panel (c) is the collision average tilting, exactly as §3 describes.
- 4. And the loser pays, in dB.** At $\lambda = 0.01$, B loses 4.26 dB to collisions while A loses 0.64 — 6.6 times as much, for a region that touches those rows equally often. Panel (e): what B pays grows $0.92 \rightarrow 1.85 \rightarrow 4.26$ dB as its weight falls, while A’s cost *shrinks* $0.89 \rightarrow 0.72 \rightarrow 0.64$. The shared capacity is not destroyed by the collision; it is transferred.
- 5. The transfer is legible row by row.** Panel (f) zeroes shared rows grouped by A’s share of their gradient mass and asks who notices. At $\lambda = 1$ the answer tracks the diagonal: the 0–0.4 bin (2,549 rows, mass mostly from B) hands A only 25% of the damage, and the 0.9–1 bin (387 rows) hands A 99%. At equal loss weights nothing imposed that split — it is what the residuals happened to do — and it still predicts, row by row, whose picture the row is holding.

What this does not show

- **The loser is not unharmed, and A is not unharmed either.** At $\lambda = 0.01$ the dominant bin holds 10,094 rows and zeroing them still costs B 7.48 dB. Panel (f) sits *below* the diagonal at low λ : B reads those rows constantly, it just no longer sets them. “Reflects the needs of the higher-weighted point” is a statement about where the row settles, not a claim that the other point stopped depending on it.
- **Importance here is a loss weight**, imposed from outside. In a radiance field it is density $\times$ visibility, which the fit itself produces and which therefore co-evolves with the table. That feedback is not present in this test.
- **This is a 2D image under an L2 loss with uniform sampling.** The sampling density of the two regions is equal by construction; a task whose samples concentrate somewhere — the paper’s “online adaptivity” — is a different measurement.
- **The dense control has 754,783 rows against 12,692**, so its absolute PSNR is not comparable with the colliding runs. Every claim above is either a within-run A-versus-B comparison or a same-region difference across the two tables, both of which are unaffected.

Reproduce

```
python scripts/collision_audit.py          # 6 fits, writes the png and the json
python scripts/collision_audit.py --replot  # rebuild the figure from the json
```

Every number in this note is read from `figures/collision_audit.json`, and this note is generated by `note/build_note.py` from that file.